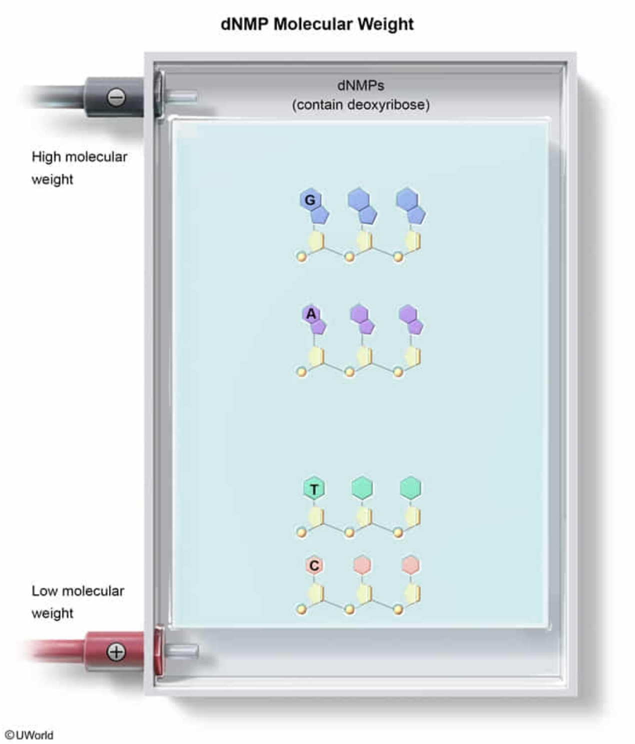
A single-stranded DNA oligonucleotide composed of which of the following would move most slowly down an alkaline agarose gel during electrophoresis?

- A. Deoxyadenosine monophosphate [19%]
- B. Deoxythymidine monophosphate [13%]
- C. Deoxycytidine monophosphate [12%]
- ✔ D. Deoxyguanosine monophosphate [54%]

Correct
55% answered correctly

Explanation:



Gel electrophoresis **separates molecules by molecular weight**, with larger molecules migrating more slowly than smaller molecules. DNA is composed of the four **deoxyribonucleotides (dNMPs)**, each of which has a different molecular weight. The purine deoxyguanosine monophosphate (dGMP) is the largest dNMP, followed by deoxyadenosine monophosphate (dAMP), deoxythymidine monophosphate (dTMP), and deoxycytidine monophosphate (dCMP).

An oligonucleotide is a short strand of DNA, and its molecular weight is determined by its composition of dNMPs. An oligonucleotide composed entirely of dGMP will be larger than any other oligonucleotide of the same length and therefore would move the most slowly through an agarose gel.

(Choices A, B, and C) Deoxyguanosine has the largest molecular weight of any nucleotide, so all other nucleotides will move faster through an agarose gel.

Educational objective:
DNA is a polymer composed entirely of deoxyribonucleoside monophosphate (dNMP) monomers. Ribonucleotides are not incorporated into replicated DNA. The molecular weight of dNMPs in decreasing order is deoxyguanosine (dGMP), deoxyadenosine (dAMP), deoxythymidine (dTMP), and deoxycytidine (dCMP).

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